Type MS · Least or greatest number of terms for an arithmetic sum

Answer 1 of 4 · Question 2 part whole · 7 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 3 · find part whole in this mark scheme extract

Number		
2	$a+2d=8 \qquad a+4d=20$	B1
	$2d=12 \Rightarrow d=$	M1
	$(a=)-4 \qquad (d=)6$	A1
	$S_n=\frac{n}{2}(2\times"-4"+(n-1)"6")(>220)$	M1
	$\left(\frac{n}{2}(-8+6n-6)>220\right)$ or $3n^2-7>220$ oe	M1
	Solution using 3TQ:	
	$6n^2-14n-440>0$ or $3n^2-7n-220>0$ oe	
	$n=\frac{7\pm\sqrt{(-7)^2-4(3)(-220)}}{2\times3} \Rightarrow n=(9.80925,-7.47592)$	M1
	$n=10$	A1
	Solution using trial and improvement:	[7]
	$n=9:\ 3\times(9)^2-7\times(9)=180<220$	[M1]
	$n=10:\ 3\times(10)^2-7\times(10)=230>220$	
	Hence minimum value is $n=10$	[A1]
		Total 7 marks

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Answer 1 of 4 · Question 2 part whole · 7 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 3 · find part whole in this mark scheme extract

Part	Mark	Additional Guidance
2	B1	For both correct equations
	M1	For a complete attempt to solve their 2 linear simultaneous equations, allow one processing error.
	A1	For both correct: $(a=)-4$ and $(d=)6$. Values may be implied by subsequent working.
	ALT for first 3 marks: Uses the third term	
	B1	Let the 3rd term be T $20-T=d$ $T-8=d$
	M1	Solves the above equations simultaneously by any method to find d **AND** finds a value for a
	A1	For both correct $a=-4$ and $d=6$. Values may be implied by subsequent working
	M1	Correct substitution of their values for a and d into the correct formula for S_n (doesn't need to be placed > 220 at this point)
	M1	For the correct inequality shown with their values oe
	M1	For a full and correct method to solve their 3TQ leading to at least one value for n. Award this mark for the positive root [9.809…] without explicit working seen [use of calculator] following a correct 3TQ seen. If the 3TQ is incorrect and there is no evidence of explicit working to solve their 3TQ do not award this mark.
	A1	For choosing only positive root, and rounding to $n = 10$ If a candidate offers a positive and negative value without rejecting the negative [-7.47…] do not award this mark.
	ALT: uses T & I Final two marks	
	M1	Attempts $n = 10$ **AND** $n = 9$ When $n = 10$ $S = 230$ and when $n = 9$ $S = 180$ **These values must be seen**
	A1	For choosing $n = 10$

Type MS · Least or greatest number of terms for an arithmetic sum

Answer 1 of 4 · Question 2 part whole · 7 marks

2025 JUN · Paper 01 (regional R) · extract 3 of 3 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Least or greatest number of terms for an arithmetic sum

Answer 2 of 4 · Question 6 part b · 3 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number		
6 (a)	$a = 4\times1-3=1, \quad (d=4)$ $\sum_{r=1}^{n} 4r-3 = \frac{n}{2}\left(2\times1+(n-1)4\right) = n(2n-1)^*$	B1 M1A1 [3]
(b)	$n(2n-1)>1000 \Rightarrow 2n^2-n-1000>0$ $\frac{-(-1)\pm\sqrt{(-1)^2-4\times2\times(-1000)}}{2\times2} \Rightarrow n>22.612... \Rightarrow n=23$	M1 M1A1 [3]
(c)	$3t_{(n+7)}+18 = S_{(n+4)}$ $\Rightarrow 3\left[4(n+7)-3\right]+18=(n+4)\left[2(n+4)-1\right]$ $\Rightarrow 2n^2+3n-65=0$ $2n^2+3n-65=(2n+13)(n-5)=0 \Rightarrow n=5$	 M1 A1 depM1A1 [4]
		Total 10 marks

Type MS · Least or greatest number of terms for an arithmetic sum

Answer 2 of 4 · Question 6 part b · 3 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Least or greatest number of terms for an arithmetic sum

Answer 3 of 4 · Question 4 part b.ii · 3 marks

2025 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

4(a)	$\sum_{r=1}^{n}(4r+A)=\frac{n}{2}\left(2(4+A)+(n-1)4\right)=\frac{n}{2}(4+2A+4n)$ Then $\frac{n}{2}(4+2A+4n)=\frac{n}{2}(4n-6)$ $4+2A=-6\Rightarrow A=-5^*$	M1 M1 A1 cso [3]
(b) (i)	$\sum_{21}^{60}(4r-5)=\frac{60}{2}(4\times60-6)-\frac{20}{2}(4\times20-6)=[7020-740]=6280$ **ALT 1** $U_{21}=4\times21-5=79 \quad U_{60}=4\times60-5=235$ $\sum_{21}^{60}(4r-5)=\frac{40}{2}(79+235)=6280$ **ALT 2** $S_{40}=\frac{40}{2}\left(2(79)+(40-1)\times4\right)$ $=20\times(158+156)=6280$	M1A1 [2] [M1A1] [M1A1]
(ii)	$\sum_{r=1}^{n}(4r-5)<3418\Rightarrow\frac{n}{2}(4n-6)<3418\Rightarrow4n^2-6n-6836<0$ $\Rightarrow 4n^2-6n-6836<0\Rightarrow n<42.096....$ $\Rightarrow n=42$	M1 M1 A1 [3]
		Total 8 marks

Type MS · Least or greatest number of terms for an arithmetic sum

Answer 3 of 4 · Question 4 part b.ii · 3 marks

2025 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question	Scheme	Marks

Type MS · Least or greatest number of terms for an arithmetic sum

Answer 4 of 4 · Question 3 part whole · 4 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

3	$S_n < -450 \Rightarrow \frac{n}{2}\left(2\times 16+[n-1](-5)\right) < -450$ $\Rightarrow 37n-5n^2 < -900 \Rightarrow 5n^2-37n-900>0$	M1A1
	$n=\frac{-(-37)\pm\sqrt{(-37)^2-4\times 5\times(-900)}}{2\times 5} \Rightarrow n=17.617..$ so $n=18$	M1A1 [4]
		Total 4 marks

Question	Notes	Marks
3	Uses the correct summation formula and sets <, > or = to –450 $S_n < -450 \Rightarrow \frac{n}{2}\left(2\times 16+[n-1](-5)\right) < -450$	M1
	Forms a correct 3TQ with their expression $37n-5n^2 < -900 \Rightarrow 5n^2-37n-900>0$ Accept <, > or = 0 and accept terms in any order.	A1
	Attempts to solve their 3TQ using a valid method. [See General Guidance} $n=\frac{-(-37)\pm\sqrt{(-37)^2-4\times 5\times(-900)}}{2\times 5} \Rightarrow n=...$	M1
	$n=17.617...$ so $n=18$ [Other root is – 10.217….]	A1
		Total 4 marks

Type MS · Least or greatest number of terms for an arithmetic sum

Answer 4 of 4 · Question 3 part whole · 4 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question	Scheme	Marks